Appendix B. Simple Connectivity and Jordan Curve Theorem

Equivalent forms of simple connectivity, the Jordan curve theorem, general Cauchy theorem

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Overview

Simple connectivity underlies many basic results of complex analysis. Recall (Chapter 3) that a region Ω is *simply connected* if any two curves in Ω with the same end-points are homotopic; from this we deduced that

$$\int_\eta f(\zeta)\,d\zeta = 0 \tag{1}$$

for every closed curve $\eta \subset \Omega$ and every f holomorphic in Ω. In this appendix we prove:

(I) Ω is simply connected if and only if it is *holomorphically simply connected*, i.e. (1) holds for all closed $\eta \subset \Omega$ and all f holomorphic in Ω.

(II) A bounded region is simply connected if and only if its complement is connected.

(III) Ω is simply connected if and only if no curve in Ω winds around a point of Ω^c.

In the second part we answer: given a simple closed curve γ, what is the region "enclosed by γ"? We prove the Jordan curve theorem for piecewise-smooth curves: γ^c has exactly two components, the bounded one is simply connected and both have boundary γ. This yields a general form of Cauchy's theorem extending the version for toy contours of Chapter 2. As in Chapter 1, "curve" means "piecewise-smooth curve" unless stated otherwise.

1 Equivalent formulations of simple connectivity

Theorem 1.1

A region Ω is holomorphically simply connected if and only if Ω is simply connected.

Proof. If Ω is simply connected, it is holomorphically simply connected by Corollary 5.3. Conversely, let Ω be holomorphically simply connected. If $\Omega = \mathbb{C}$ it is simply connected. Otherwise the proof of the Riemann mapping theorem (Theorem 3.1) applies verbatim: the only use of simple connectivity there is to construct a holomorphic logarithm (hence square root) of a non-vanishing holomorphic function g, namely $\log g(z) = c + \int_{z_0}^z g'/g$, which is well defined precisely because (1) holds. Hence Ω is conformally equivalent to $\mathbb{D}$. Since $\mathbb{D}$ is simply connected and a conformal map carries homotopies to homotopies, Ω is simply connected. ∎

Theorem 1.2

If Ω is a bounded region in $\mathbb{C}$, then Ω is simply connected if and only if Ω^c is connected.

Boundedness is needed: an infinite strip is simply connected but its complement has two components. If complements are taken in the Riemann sphere, the conclusion holds for all regions.

Proof that Ω^c connected $\Rightarrow$ Ω simply connected. By Theorem 1.1 it suffices to show $\int_\eta f = 0$ for $\eta \subset \Omega$ closed and f holomorphic in Ω. With $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$, let

$$K_\delta = \{z \in \overline{\Omega} : d(z, \Omega^c) \geq \delta\}.$$

K_δ is compact (Ω is bounded), $K_\delta \subset \Omega$, and $\eta \subset K_\delta$ for small δ since η is compact. We show that K_δ^c is connected.

Suppose $K_\delta^c = O_1 \cup O_2$ with O_1, O_2 disjoint, open, non-empty, and set $F_i = O_i \cap \Omega^c$. Since $\Omega^c \subset K_\delta^c$, $\Omega^c = F_1 \cup F_2$ with $F_1 \cap F_2 = \emptyset$.

Step F_1 is closed. Let $z_n \in F_1$, $z_n \to z$. Then $z \in \Omega^c$ (Ω^c is closed), so $z \in O_1 \cup O_2$. If $z \in O_2$, then $z_n \in O_2$ for large n as O_2 is open, contradicting $z_n \in O_1$. So $z \in O_1$, i.e. $z \in F_1$. Likewise F_2 is closed.

Step $F_1 \neq \emptyset$. Otherwise $\Omega^c \subset O_2$. Pick $w \in O_1$. Since $w \notin K_\delta$, there is $z \in \Omega^c$ with $|w - z| < \delta$. Every point of the segment $[w, z]$ is within distance $< \delta$ of $z \in \Omega^c$, hence lies in K_δ^c. So the connected segment lies in $O_1 \cup O_2$ and meets both ($w \in O_1$, $z \in O_2$), a contradiction. Likewise $F_2 \neq \emptyset$.

Thus Ω^c is disconnected, contrary to hypothesis; so K_δ^c is connected. By Runge's theorem (Theorem 5.7) there are polynomials $P_n \to f$ uniformly on K_δ, hence on η. Since $\int_\eta P_n = 0$ (polynomials have primitives), $\int_\eta f = \lim \int_\eta P_n = 0$. ∎

The converse is proved after Theorem 1.4, using winding numbers.

Winding numbers

If γ is closed and $z \notin \gamma$, each loop of γ around z changes $\frac{1}{2\pi}\arg(\zeta - z)$ by ± 1. Since $\log w = \log|w| + i \arg w$, the quantity $\frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}$ should count these loops.

Definition (Winding number)

The **winding number** (or **index**) of a closed curve γ around $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}.$$

For $\gamma(t) = e^{ikt}$, $0 \le t \le 2\pi$ ($k \in \mathbb{N}$), $W_\gamma(z) = k$ for $|z| < 1$ and 0 for $|z| > 1$; for e^{-ikt} the values are $-k$ and 0. For a positively oriented toy contour, $W_\gamma = 1$ in its interior and 0 in its exterior.

Lemma 1.3

Let γ be a closed curve in $\mathbb{C}$.

(i) If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.

(ii) If z, w lie in the same connected component of γ^c, then $W_\gamma(z) = W_\gamma(w)$.

(iii) If z lies in the unbounded component of γ^c, then $W_\gamma(z) = 0$.

Proof. (i) Let $\gamma : [0,1] \to \mathbb{C}$ parametrize the curve and $G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z}\, ds$. G is continuous and, except at finitely many points, differentiable with $G' = \gamma'/(\gamma - z)$. Then $H(t) = (\gamma(t) - z)e^{-G(t)}$ is continuous with $H' = \gamma' e^{-G} - (\gamma - z)G'e^{-G} = 0$ except at finitely many points, so H is constant: $H(1) = H(0) = \gamma(0) - z$. As $\gamma(1) = \gamma(0) \neq z$, $e^{-G(1)} = 1$, so $G(1) = 2\pi i\, W_\gamma(z) \in 2\pi i\mathbb{Z}$.

(ii) W_γ is a continuous function of $z \in \gamma^c$ (the integrand depends continuously on z, uniformly on γ), integer-valued by (i), hence constant on each connected component.

(iii) $|W_\gamma(z)| \le \frac{\ell(\gamma)}{2\pi\, d(z,\gamma)} \to 0$ as $|z| \to \infty$; since W_γ is constant on the unbounded component by (ii), it is 0 there. ∎

Theorem 1.4

A bounded region Ω is simply connected if and only if $W_\gamma(z) = 0$ for every closed curve $\gamma \subset \Omega$ and every $z \notin \Omega$.

Proof. If Ω is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in Ω, and Cauchy's theorem gives $W_\gamma(z) = 0$.

Conversely, by the part of Theorem 1.2 already proved, it suffices to show Ω^c is connected. Suppose not: $\Omega^c = F_1 \cup F_2$ with F_1, F_2 disjoint, closed, non-empty. Only one of them can be unbounded (Ω is bounded, so Ω^c contains the connected set $\{|z| > R\}$), so we may assume F_1 is bounded, hence compact. We construct a closed curve $\gamma \subset \Omega$ and $w \in F_1$ with $W_\gamma(w) \neq 0$, using the following

lemma. ■

Lemma 1.5

Under the above assumptions, let $w \in F_1$. There is a finite collection $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ of closed squares of a uniform grid $\mathcal{G}$ such that:

(i) w is in the interior of Q_1;

(ii) the interiors of Q_j and Q_k are disjoint for $j \neq k$;

(iii) F_1 is contained in the interior of $\bigcup_j Q_j$;

(iv) $\bigcup_j Q_j$ is disjoint from F_2;

(v) the boundary of $\bigcup_j Q_j$ lies in Ω and is a finite union of disjoint simple closed polygonal curves.

Proof of Theorem 1.4, assuming Lemma 1.5. Orient each ∂Q_j positively. Since w is interior to Q_1 and $w \notin Q_j$ for $j > 1$,

$$\sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1. \tag{2}$$

A side shared by two squares of $\mathcal{Q}$ is traversed twice in opposite directions and cancels. The remaining sides, oriented with $\bigcup Q_j$ on the left, form the polygons $\gamma_1, \dots, \gamma_M$ of (v), so $\sum_{j=1}^{M} W_{\gamma_j}(w) = 1$. Hence $W_{\gamma_{j_0}}(w) \neq 0$ for some j_0, while $\gamma_{j_0} \subset \Omega$ and $w \notin \Omega$: a contradiction. ■

Proof of Lemma 1.5. F_1 is compact and F_2 closed, so $d = d(F_1, F_2) > 0$. Take a grid $\mathcal{G}_0$ of closed squares of side $\delta < d/100$ such that w is the center of a square R_1, and let $\mathcal{R} = \{R_1, \dots, R_m\}$ be the squares of $\mathcal{G}_0$ meeting F_1.

Step (i)–(iv) and $\partial \bigcup R_j \subset \Omega$. (i), (ii) are clear. Every point of $\bigcup R_j$ is within $\sqrt{2}\delta < d$ of F_1, giving (iv). If $z \in F_1$, every square of $\mathcal{G}_0$ containing z meets F_1 and so belongs to $\mathcal{R}$; these squares cover a neighborhood of z, giving (iii). A *boundary side* (a side of exactly one square of $\mathcal{R}$) is a side of a square not in $\mathcal{R}$, so it misses F_1; it also misses F_2 by (iv). Hence $\partial \bigcup R_j \subset \Omega$.

Step Bad vertices. The boundary of $\bigcup R_j$ is the union of the boundary sides; their end-points are boundary vertices. At a grid vertex P the four surrounding squares give 0, 2 or 4 boundary sides at P, and 4 occurs exactly when two diagonally opposite squares are in $\mathcal{R}$ and the other two are not. Call such a P *bad*.

Step Refinement. Let $\mathcal{G}$ be the grid obtained by cutting each square of $\mathcal{G}_0$ into 9 equal subsquares, and take all $9m$ subsquares of squares of $\mathcal{R}$, with Q_1 the middle subsquare of R_1 (so w is its center). At each bad vertex P add the two subsquares of $\mathcal{G}$ having P as a corner and lying in the two missing squares of $\mathcal{G}_0$. Call the result $\mathcal{Q} = \{Q_1, \dots, Q_n\}$. Now all four small squares at P are in $\mathcal{Q}$, so P is interior. An added subsquare s at P has three other corners. At the two lying on sides of $\mathcal{G}_0$, three of the four surrounding small squares are in $\mathcal{Q}$. At the one inside a big square, only s is. So none of them is bad. Added squares at different bad vertices sit at different corners of a big square, so they do not touch. Hence $\mathcal{Q}$ has no bad vertices. (i)–(iv) persist: the added squares lie in squares of $\mathcal{G}_0$ adjacent to $\mathcal{R}$, within $3\delta < d$ of F_1. The new boundary sides lie in squares of $\mathcal{G}_0$ not in $\mathcal{R}$, so they miss F_1, and also F_2. So $\partial \bigcup Q_j \subset \Omega$.

Step (v). Every boundary vertex of $\bigcup Q_j$ now meets exactly two boundary sides. With the orientation leaving $\bigcup Q_j$ on the left, one of them enters the vertex and the other leaves it (check the cases of one, two adjacent, or three surrounding squares). Starting from a boundary side $[a_1, a_2]$ we get a unique continuation $[a_2, a_3], [a_3, a_4], \dots$. There are finitely many sides, so some vertex repeats; let a_m be the first repetition, $a_m = a_k$ with $k < m$. If $k > 1$, then a_k meets the three distinct sides $[a_{k-1}, a_k]$, $[a_k, a_{k+1}]$, $[a_{m-1}, a_m]$, which is impossible. So $k = 1$ and $a_1 \cdots a_m$ is a simple closed polygon. Removing its sides and repeating exhausts the boundary sides, giving finitely many disjoint simple closed polygons (disjoint because each vertex lies on only one of them). ■

Proof of Theorem 1.2, converse. Let Ω be bounded and simply connected. If Ω^c were disconnected, the proof of Theorem 1.4 produces a closed curve $\gamma \subset \Omega$ and $w \notin \Omega$ with $W_\gamma(w) \neq 0$. This contradicts the first part of Theorem 1.4, since Ω is simply connected. ■

2 The Jordan curve theorem

The curves in the statements are piecewise-smooth, but the proofs also use curves that are merely continuous (the curves γ_η below).

Theorem 2.1 (Jordan arc theorem)

Let γ be a simple (non-closed) piecewise-smooth curve in the plane. Then γ^c is an open connected set whose boundary is precisely γ.

Theorem 2.2 (Jordan curve theorem)

Let γ be a simple closed piecewise-smooth curve in the plane. Then γ^c consists of two disjoint connected open sets. Exactly one of them is bounded and simply connected; it is the **interior** Ω_γ of γ. The other is unbounded; it is the **exterior** $\mathcal{U}_\gamma$. With the appropriate orientation of γ,

$$W_\gamma(z) = \begin{cases} 1 & z \in \Omega_\gamma, \\ 0 & z \in \mathcal{U}_\gamma. \end{cases}$$

Remark

Both theorems hold for merely continuous curves, but the proofs are harder; the piecewise-smooth case suffices for many applications.

Theorem 2.3

If f is holomorphic in the interior Ω_γ of a simple closed curve γ, then $\int_\eta f(\zeta)\,d\zeta = 0$ for every closed curve $\eta \subset \Omega_\gamma$.

Proof (supplied). Ω_γ is a simply connected region by Theorem 2.2, so this is Cauchy's theorem, Corollary 5.3. ■

Idea of the proof of Theorem 2.1. γ^c is open, so it suffices to show it is pathwise connected. Join $z, w \notin \gamma$ by a curve. If it meets γ, first go from z to a point z' and from w to a point w' near γ without meeting γ. Then join z' to w' by traveling "parallel" to γ, going around an end-point if necessary. If γ parametrizes a smooth piece, $\gamma'(t) \neq 0$ is tangent to it, $i\gamma'(t)$ is normal, and $\gamma(t) + i\eta\gamma'(t)$ is a "parallel" curve.

In the next three lemmas and two propositions, γ_0 is a *simple smooth* curve with arc-length parametrization $\gamma : [0, L] \to \mathbb{C}$, $|\gamma'(t)| = 1$ (every smooth curve has one). An *interior point* of γ_0 is a point $\gamma(t)$ with $0 < t < L$ (not to be confused with the interior of Theorem 2.2).

Lemma 2.4

For $\eta \in \mathbb{R}$ let γ_η be the continuous curve $\gamma_\eta(t) = \gamma(t) + i\eta\gamma'(t)$, $0 \le t \le L$. There is $\eta_1 > 0$ such that $\gamma_0 \cap \gamma_\eta = \emptyset$ whenever $0 < |\eta| < \eta_1$.

Proof.

Step Local version. For $s, t \in [0, L]$,

$$\gamma_\eta(t) - \gamma(s) = \int_s^t \gamma'(u)\,du + i\eta\gamma'(t) = \int_s^t [\gamma'(u) - \gamma'(t)]\,du + (t - s + i\eta)\gamma'(t).$$

By uniform continuity of γ' there is $\delta > 0$ with $|\gamma'(x) - \gamma'(y)| < 1/2$ for $|x - y| < \delta$. If $|s - t| < \delta$, then using $|\gamma'(t)| = 1$ and $2|a + ib| \geq |a| + |b|$,

$$|\gamma_\eta(t) - \gamma(s)| \geq |t - s + i\eta| - \frac{|t - s|}{2} \geq \frac{|\eta|}{2}.$$

So $\gamma_\eta(t) \neq \gamma(s)$ when $|t - s| < \delta$ and $\eta \neq 0$.

Step Global version. Choose $0 = t_0 < \cdots < t_n = L$ with $t_{k+1} - t_k < \delta/2$, and set $I_k = \{t : |t - t_k| \leq \delta/4\}$, $J_k = \{t : |t - t_k| < \delta/2\}$, $J_k' = \{t \in [0, L] : |t - t_k| \geq \delta/2\}$ (intersected with $[0, L]$). The I_k cover $[0, L]$. If $t \in I_k$ and $s \in J_k$ then $|t - s| < \delta$, so

$$\gamma_\eta(I_k) \cap \gamma(J_k) = \emptyset \qquad (\eta \neq 0). \tag{3}$$

Since γ_0 is simple, the compact sets $\gamma(I_k)$ and $\gamma(J_k')$ are disjoint, at distance $d_k > 0$. For $t \in I_k$, $s \in J_k'$,

$$|\gamma_\eta(t) - \gamma(s)| \geq |\gamma(t) - \gamma(s)| - |\eta| \geq d_k - |\eta| \geq d_k/2 \quad \text{if } |\eta| \leq d_k/2,$$

so

$$\gamma_\eta(I_k) \cap \gamma(J_k') = \emptyset \qquad (|\eta| < d_k/2). \tag{4}$$

With $\eta_1 = \min_k d_k/2$, (3) and (4) give $\gamma_0 \cap \gamma_\eta = \emptyset$ for $0 < |\eta| < \eta_1$. ■

Lemma 2.5

Suppose $z \notin \gamma_0$ and $z_0 \in \gamma_0$ is a point closest to z with $z_0 = \gamma(t_0)$, $0 < t_0 < L$ (this holds if z is closer to an interior point of γ_0 than to either end-point). Then $z = \gamma(t_0) + i\eta\gamma'(t_0)$ for some real $\eta \neq 0$, i.e. $z \in \gamma_\eta$.

Proof. Near t_0, $z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|)$, so

$$|z - z_0|^2 \leq |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\,\mathrm{Re}\,[(z - \gamma(t_0))\overline{\gamma'(t_0)}] + o(|t - t_0|).$$

Since $t - t_0$ takes both signs, $\mathrm{Re}[(z - \gamma(t_0))\overline{\gamma'(t_0)}] = 0$, i.e. $(z - \gamma(t_0))\overline{\gamma'(t_0)} = i\eta$ with $\eta \in \mathbb{R}$. As $|\gamma'(t_0)| = 1$, $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, so $z - \gamma(t_0) = i\eta\gamma'(t_0)$; and $\eta \neq 0$ because $z \notin \gamma_0$. ■

If $z \in \gamma_\eta$ and $w \in \gamma_\epsilon$ ($\eta, \epsilon \neq 0$) are obtained as in Lemma 2.5, we say z, w are on the **same side** of γ_0 if η, ϵ have the same sign, and on **opposite sides** otherwise. We do not define "the two sides of γ_0"; the notion refers to the chosen representations, and we do not claim the two cases are exclusive. Points on the same side can be joined almost directly by a curve parallel to γ_0. Points on opposite sides require going around an end-point.

Proposition 2.6

Let A, B be the end-points of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A, B\}$ (or a subset thereof). If $z, w \notin \gamma_0$ lie on the same side of γ_0, and each is closer to an interior point of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous curve in the complement of $K \cup \gamma_0$.

Proof. By Lemma 2.5, with $z_0 = \gamma(t_0)$, $w_0 = \gamma(s_0)$ interior points closest to z, w,

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) + i\epsilon_0\gamma'(s_0),$$

with η_0, ϵ_0 of the same sign, say positive, and say $t_0 \leq s_0$.

Step Normal segments. The half-open segment from z to z_0 lies in $(K \cup \gamma_0)^c$. If p is on it with $p \neq z_0$ and $p \in K \cup \gamma_0$, then $|z-p| < |z-z_0| = d(z, K \cup \gamma_0)$, which is absurd. So for all small $\eta > 0$, z can be joined to $z_\eta = \gamma(t_0) + i\eta\gamma'(t_0)$, and likewise w to $w_\eta = \gamma(s_0) + i\eta\gamma'(s_0)$.

Step Parallel curve. Choose $\eta < \eta_1$ (Lemma 2.4) and η smaller than the distance from K to $\{\gamma(t) : t_0 \le t \le s_0\}$. This distance is positive since this arc avoids A, B. Then $\{\gamma_\eta(t) : t_0 \le t \le s_0\}$ joins z_η to w_η. It misses γ_0 by Lemma 2.4, and it misses K since $|\gamma_\eta(t) - \gamma(t)| = \eta$. ■

Lemma 2.7

There is $\eta_2 > 0$ such that the half-neighborhood

$$N = \{\gamma(L) + \eta e^{i\theta}\gamma'(L) : -\pi/2 \le \theta \le \pi/2,\ 0 < \eta < \eta_2\}$$

is disjoint from γ_0.

Proof. As in Lemma 2.4,

$$\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)]\,du + (L - t + \eta e^{i\theta})\gamma'(L).$$

Choose δ with $|\gamma'(u) - \gamma'(L)| < 1/2$ for $|u - L| < \delta$. Since $\cos\theta \ge 0$, $|L - t + \eta e^{i\theta}|^2 \ge (L-t)^2 + \eta^2 \ge \frac{1}{2}(L - t + \eta)^2$. So for $L - \delta < t \le L$,

$$|\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t)| \ge \frac{L - t + \eta}{\sqrt{2}} - \frac{L-t}{2} \ge \frac{\eta}{\sqrt{2}} > 0.$$

Thus $\gamma(t) \notin N$ for $L - \delta < t \le L$. Finally, take η_2 less than the distance from $\gamma(L)$ to $\{\gamma(t) : 0 \le t \le L - \delta\}$, which is positive since γ_0 is simple. ■

Proposition 2.8

Let A be an end-point of γ_0 and K a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A\}$. If $z, w \notin \gamma_0$ are closer to interior points of γ_0 than to K or to the end-points of γ_0, then z and w can be joined by a continuous curve in the complement of $\gamma_0 \cup K$.

Proof. If z, w are on the same side, apply Proposition 2.6. So let them be on opposite sides, and (reversing the parametrization if needed) $A = \gamma(0)$, so the other end-point $B = \gamma(L) \notin K$. Write

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) - i\epsilon_0\gamma'(s_0), \qquad \eta_0, \epsilon_0 > 0,$$

and let $\tau = \min(t_0, s_0) > 0$. Choose $\eta > 0$ smaller than η_1, η_2 (Lemmas 2.4, 2.7), and the distance from K to $\{\gamma(t) : \tau \le t \le L\}$ (positive since this arc avoids A). As in Proposition 2.6, join z to $\gamma_\eta(t_0)$ by a normal segment. Then follow γ_η from t_0 to L to reach $\gamma(L) + i\eta\gamma'(L)$. Then follow the half-circle $\gamma(L) + \eta e^{i\theta}\gamma'(L)$, θ from $\pi/2$ down to $-\pi/2$, inside N, to reach $\gamma(L) - i\eta\gamma'(L)$. Then follow $\gamma_{-\eta}$ from L back to s_0, and the normal segment to w. The parallel pieces miss γ_0 by Lemma 2.4; the half-circle misses γ_0 by Lemma 2.7. All pieces lie within η of $\{\gamma(t) : \tau \le t \le L\}$, so they miss K. ■

Proof of Theorem 2.1.

Step Boundary. $O = \gamma^c$ is open and $\partial O \subset \gamma$. Every point of a smooth piece of γ is a limit of points of O: by Lemma 2.4 (applied to a small sub-arc, and noting the rest of γ is at positive distance), $\gamma(t) + i\eta\gamma'(t) \notin \gamma$ for small $\eta \neq 0$. Since ∂O is closed and the smooth points are dense in γ, $\partial O = \gamma$.

Step Connectedness: induction on the number of smooth pieces. First let γ be smooth, with end-points A, B, and let $Z, W \notin \gamma$. Join Z to W by a polygonal path Λ avoiding A and B. If Λ misses γ we are done. Otherwise let p and q be the first and last points of Λ on γ; they are interior points of γ. Points z of Λ just before p satisfy $d(z, \gamma) \le |z - p| \to 0$, while $|z - A|, |z - B|$ stay bounded

below. So we may take such a $z \notin \gamma$ closer to an interior point of γ than to A, B, joined to Z along Λ without meeting γ. Similarly take w just after q. Proposition 2.8 with $K = \emptyset$ joins z to w in γ^c.

Now suppose the result holds for curves of $n-1$ smooth pieces, and let $\gamma = K \cup \gamma_0$ with K the union of $n-1$ consecutive pieces and γ_0 smooth. Then K is compact and $K \cap \gamma_0$ is a single end-point A of γ_0. Given $Z, W \notin \gamma$, the induction hypothesis gives a curve from Z to W missing K. Since K^c is open and connected, we may take it polygonal and avoiding the other end-point of γ_0. If it meets γ_0, it does so at interior points. As above, take z (resp. w) just before the first (resp. after the last) meeting point; it is closer to an interior point of γ_0 than to K or to the end-points, since its distance to K is bounded below. Proposition 2.8 joins z to w in $(\gamma_0 \cup K)^c = \gamma^c$. ∎

Proof of Theorem 2.2.

Step At most two components. Fix W outside a large disc containing γ, and let $\mathcal{U}$ be the set of points joined to W by a continuous curve in γ^c. $\mathcal{U}$ is open and connected, and contains the exterior of the disc. Let $\Omega = \gamma^c - \mathcal{U}$; we show Ω is connected. Remove a smooth piece γ_0 (with end-points A, B) from γ, and let K be the remaining arc, so $K \cap \gamma_0 = \{A, B\}$. For $Z \in \Omega$, Theorem 2.1 applied to K gives a curve Λ_Z from Z to W missing K. Since $Z \notin \mathcal{U}$, Λ_Z meets γ_0, necessarily at interior points. As in the proof of Theorem 2.1, choose $z, w \in \Lambda_Z$ closer to interior points of γ_0 than to K or to A, B, such that the pieces of Λ_Z from Z to z and from w to W lie in γ^c. Then z, w are on opposite sides of γ_0. Otherwise Proposition 2.6 would join z to w in γ^c, hence Z to W, contradicting $Z \notin \mathcal{U}$. If $Z_1 \in \Omega$ is another point, with corresponding z_1, w_1 on opposite sides, then z and z_1 are on the same side. Otherwise z_1 is on the same side as w, and Proposition 2.6 joins Z_1 to W in γ^c, a contradiction. So Proposition 2.6 joins z to z_1 in γ^c, and Z, Z_1 lie in the same component. Thus γ^c has at most two components, $\mathcal{U}$ and (if non-empty) Ω.

Step Exactly two components. By Lemma 1.3 it suffices to find points with different winding numbers. Parametrize γ and fix $z_0 = \gamma(t_0)$ interior to a smooth piece, $v = \gamma'(t_0) \neq 0$. For $\epsilon > 0$ let

$$z_\epsilon = \gamma(t_0) + i\epsilon v, \qquad w_\epsilon = \gamma(t_0) - i\epsilon v.$$

For small ϵ these avoid γ, and the normal segments $\{\gamma(t_0) \pm isv : 0 < s \leq \epsilon\}$ lie in γ^c (Lemma 2.4 for the piece, plus the positive distance from $\gamma(t_0)$ to the rest of γ). By Lemma 1.3(ii),

$$\Delta = |W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)|$$

is a constant integer for all small $\epsilon > 0$. With $a(t) = \gamma(t) - \gamma(t_0)$,

$$\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} = \frac{2i\epsilon v\, \gamma'(t)}{a(t)^2 + \epsilon^2 v^2}.$$

Since $a(t) = v(t - t_0) + o(|t - t_0|)$ and $\gamma'(t) = v + o(1)$,

$$\frac{2i\epsilon v \gamma'(t)}{a(t)^2 + \epsilon^2 v^2} = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2}\,(1 + \rho(t, \epsilon)),$$

where, given $\sigma > 0$, there is $\delta > 0$ with $|\rho(t, \epsilon)| < \sigma$ for $|t - t_0| \leq \delta$, uniformly in ϵ. (Indeed $a^2 + \epsilon^2 v^2 = v^2[(t - t_0)^2 + \epsilon^2] + o((t - t_0)^2)$ and $o((t - t_0)^2)/[(t - t_0)^2 + \epsilon^2] \to 0$.) Split the integral defining $W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)$ into $|t - t_0| \geq \delta$ and $|t - t_0| < \delta$. On the first part $\gamma(t)$ stays at positive distance from $\gamma(t_0)$, while $z_\epsilon, w_\epsilon \to \gamma(t_0)$, so this part tends to 0 as $\epsilon \to 0$. On the second,

$$\frac{1}{2\pi i} \int_{|t - t_0| < \delta} \frac{2i\epsilon\, dt}{(t - t_0)^2 + \epsilon^2} = \frac{1}{\pi} \int_{-\delta/\epsilon}^{\delta/\epsilon} \frac{ds}{s^2 + 1} = \frac{2}{\pi} \arctan(\delta/\epsilon) \to 1,$$

and the ρ-term is at most $\frac{\sigma}{\pi} \int_{\mathbb{R}} \frac{ds}{s^2+1} = \sigma$ in absolute value. Letting $\epsilon \to 0$ gives $|\Delta - 1| \leq \sigma$ for every $\sigma > 0$, so $\Delta = 1$.

Hence z_ϵ and w_ϵ lie in different components, so γ^c has exactly the two components $\mathcal{U}$ and Ω. $\mathcal{U}$ is unbounded with $W_\gamma = 0$ there (Lemma 1.3(iii)), and Ω is bounded. Since $\Delta = 1$, $W_\gamma = \pm 1$ on Ω,

and after possibly reversing orientation $W_\gamma = 1$ there. Every smooth point of γ is a limit of the z_ϵ and of the w_ϵ, hence of points of both components. Smooth points are dense in γ, and the boundary of each component lies in γ. So $\gamma = \partial\Omega = \partial\mathcal{U}$. Write $\Omega_\gamma = \Omega$ and $\mathcal{U}_\gamma = \mathcal{U}$.

Step Ω_γ is simply connected. By Theorem 1.2 it suffices that $\Omega_\gamma^c = \gamma \cup \mathcal{U}_\gamma$ is connected. Suppose $\Omega_\gamma^c = F_1 \cup F_2$ with F_1, F_2 closed, disjoint, non-empty, and let $O_i = \mathcal{U}_\gamma \cap F_i$. The sets O_1, O_2 are disjoint and $\mathcal{U}_\gamma = O_1 \cup O_2$.

O_1 is open. Let $z \in O_1$; small balls B at z lie in $\mathcal{U}_\gamma \subset F_1 \cup F_2$. If every such ball met F_2, then $z \in F_2$ (F_2 closed), contradicting $F_1 \cap F_2 = \emptyset$. So some ball lies in F_1, hence in O_1. Likewise O_2 is open.

$O_1 \neq \emptyset$. Otherwise $F_1 \subset \gamma$ and $\mathcal{U}_\gamma \subset F_2$. Take $z \in F_1 \subset \gamma$. Since $\gamma = \partial\mathcal{U}_\gamma$, every ball at z meets $\mathcal{U}_\gamma \subset F_2$, so $z \in F_2$, a contradiction. Likewise $O_2 \neq \emptyset$.

This disconnects $\mathcal{U}_\gamma$, a contradiction. This completes the proof of the Jordan curve theorem for piecewise-smooth curves. ■

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9

If f is holomorphic in an open set containing a simple closed piecewise-smooth curve γ and its interior, then $\int_\gamma f = 0$.

Let $\mathcal{O} \supset \gamma \cup \Omega_\gamma$ be open with f holomorphic in $\mathcal{O}$. The idea is to build a closed curve $\Gamma \subset \Omega_\gamma$ so close to γ that $\int_\gamma f = \int_\Gamma f$. The right side vanishes by Theorem 2.3, since f is holomorphic in the simply connected set Ω_γ. Near smooth parts of γ, Γ is essentially a parallel curve γ_η (Lemma 2.4); near the junctions of smooth parts, Γ follows arcs of circles.

Lemma 2.10

Let $\gamma : [0,1] \to \mathbb{C}$ be a simple smooth curve. For all sufficiently small $\epsilon > 0$ the circle C_ϵ of center $\gamma(0)$ and radius ϵ meets γ in exactly one point.

Proof. Assume $\gamma(0) = 0$. Since $\gamma(1) \neq 0$, the intermediate value theorem applied to $|\gamma(t)|$ gives at least one intersection point for $\epsilon < |\gamma(1)|$. Suppose the claim fails: there are $\epsilon_j \to 0$ such that $|\gamma(t)| = \epsilon_j$ has two distinct solutions. Since $|\gamma(t)| \geq c(\tau) > 0$ for $t \geq \tau$ (simplicity), both solutions tend to 0. By the mean value theorem applied to $h(t) = |\gamma(t)|^2$, there are $t_j \to 0$, $t_j > 0$, with $h'(t_j) = 2\,\mathrm{Re}[\gamma'(t_j)\overline{\gamma(t_j)}] = 0$. But $\gamma(t) = \gamma'(0)t + t\varphi(t)$ and $\gamma'(t) = \gamma'(0) + \psi(t)$ with $\varphi, \psi \to 0$ as $t \to 0$, so

$$\mathrm{Re}[\gamma'(t)\overline{\gamma(t)}] = |\gamma'(0)|^2 t + o(t),$$

which is $\neq 0$ for small $t > 0$ since $\gamma'(0) \neq 0$. This is a contradiction. ■

Proof of Theorem 2.9. Choose ϵ so small that the open set U of points at distance $< \epsilon$ from γ is contained in $\mathcal{O}$. Let $P_1, \ldots, P_n$ be the consecutive junction points of the smooth parts of γ. Choose $\delta < \epsilon/10$ so small that each circle C_j centered at P_j of radius δ meets γ in exactly two points. This is possible by Lemma 2.10, applied to the two smooth parts meeting at P_j; the rest of γ is at positive distance from P_j. The two points determine two arcs of C_j, exactly one of which, $\mathcal{C}_j$, has its interior in Ω_γ. Indeed, for a smooth part with end-point P_j and small η, the curves γ_η and $\gamma_{-\eta}$ lie on opposite sides of γ, hence in different components of γ^c (proof of Theorem 2.2), and both meet C_j. The disc D_j of center P_j and radius 2δ lies in U, hence in $\mathcal{O}$.

Let B_j be the end-point of $\mathcal{C}_j$ on the smooth part γ_j from P_j to P_{j+1}, and A_{j+1} the end-point of $\mathcal{C}_{j+1}$ on γ_j. Take a chain of discs $\mathcal{D} = \{D_0, \ldots, D_K\}$ in U covering γ, with $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, and with the discs D_j among them. By Lemma 2.4 there is a continuous curve Γ_j in Ω_γ and in the

union of the discs, running parallel to γ_j, from a point near B_j on $\mathcal{C}_j$ to a point near A_{j+1} on $\mathcal{C}_{j+1}$. Since γ is only C^1, Γ_j need not be smooth; replacing it by a nearby polygonal line we may assume it is. Joining Γ_j to Γ_{j+1} along a piece of $\mathcal{C}_{j+1}$, and so on, gives a closed piecewise-smooth curve $\Gamma \subset \Omega_\gamma$.

f has a primitive on each disc of $\mathcal{D}$ (Theorem 2.1). Arguing as in the proof of Theorem 5.1, and comparing γ and Γ disc by disc, gives $\int_\gamma f = \int_\Gamma f$: on consecutive discs the primitives differ by constants, and the endpoint contributions telescope. Since $\Gamma \subset \Omega_\gamma$ and Ω_γ is simply connected, $\int_\Gamma f = 0$ by Theorem 2.3. Hence $\int_\gamma f = 0$. ∎